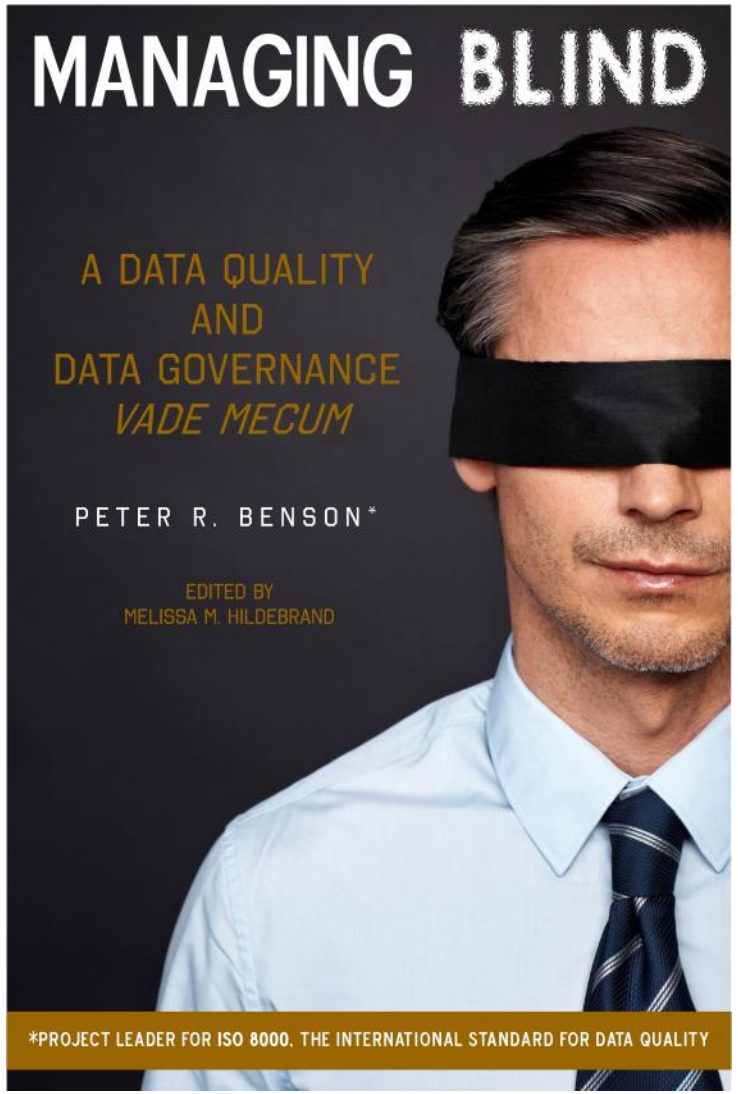


The Journey to Information Quality 2026-10-13T18:40Z

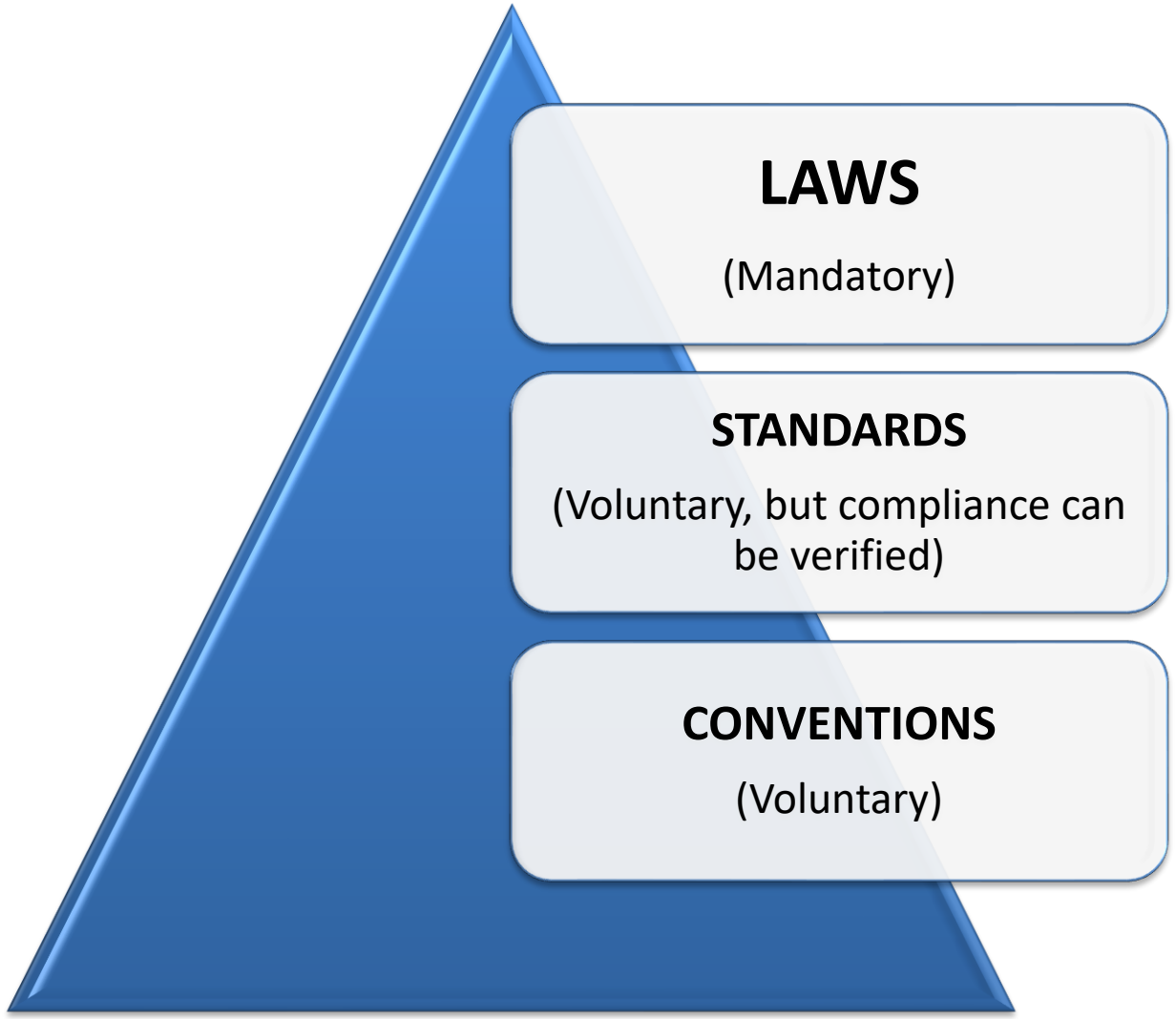
Peter R Benson | Executive Director
ISO.IBID:US-DE.BER:3031657|Code Management Association|19990420|D
Supply chain data quality and interoperability through International Standards
ECCMA | www.eccma.org | M:+1 610 462 5923 | peter.benson@eccma.net
EST/EDT (-05:00 UTC/-04:00 UTC)
https://www.linkedin.com/in/peterrichardbenson/

Incoming Chair of ISO TC 184 SC5
Interoperability, integration, and architectures for enterprise systems and automation applications
Project leader for ISO 22745, ISO 8000, ISO 25500

Agenda

- Standards
- Data and Information
- NATO – the origin of ISO 8000
- Defining quality
- Data portability
- Concept encoding and data dictionaries
- The Industrial Internet
- Formatting identifiers
- Supply chain data formatting clause
- ISO standards

Standards as Business Constraints



ISO TC 184/SC 4 Voting Members

International standards represent the "consensus" of international domain experts



- Australia, SAI; Standards Australia International, Ltd
- Austria, ON; Österreichisches Normungsinstitut
- Brazil, ABNT; Associação Brasileira de Normas Técnicas
- Bulgaria, BDS; State Agency for Standardization and Metrology
- China, SAC; Standardization Administration of China
- Czech Republic, CNI; Czech Standards Institute
- France, AFNOR; Association française de normalisation
- Germany, DIN; Deutsches Institut für Normung
- Italy, UNI; Italian National Standards Body
- Japan, JISC; Japanese Industrial Standards Committee
- Korea, KATS; Korean Agency for Technology and Standards
- Netherlands, NEN; Nederlands Normalisatie-instituut
- Norway, SN; Standards Norway
- Portugal, IPQ; Instituto Português da Qualidade
- Russia, GOST; Federal Agency on Technical Regulating and Metrology
- Saudi Arabia, SASO; The Saudi Standards, Metrology and Quality Organization
- South Africa, SABS; South African Bureau of Standards
- Spain, AENOR; Asociación Española de Normalización y Certificación
- Sweden, SIS; Swedish Standards Institute
- Switzerland, SNV; Swiss Association for Standardization
- United Kingdom, BSI; British Standards Institution
- United States, ANSI; American National Standards Institute

Standards

Standards are specifications – by claiming compliance with a standard you are publicly (and legally) announcing that you meet the requirements of the standard.

ISO Standards contain:

Scope (clause 1)

Normative references (clause 2)

- Other standards that you must conform to in order to conform to this standard

Terms and definition of terms (clause 3)

- Normative definitions are an integral part of the standard. A term may be defined in another ISO standard, in which case the standard and the definition will be clearly referenced.
- If a term is not defined in a standard, the Oxford English dictionary (OED) definition applies.

Abbreviated terms (clause 4)

Normative text (clause 5 and above)

- Also called conformance clauses. If you claim compliance you are stating that you have implemented these clauses.

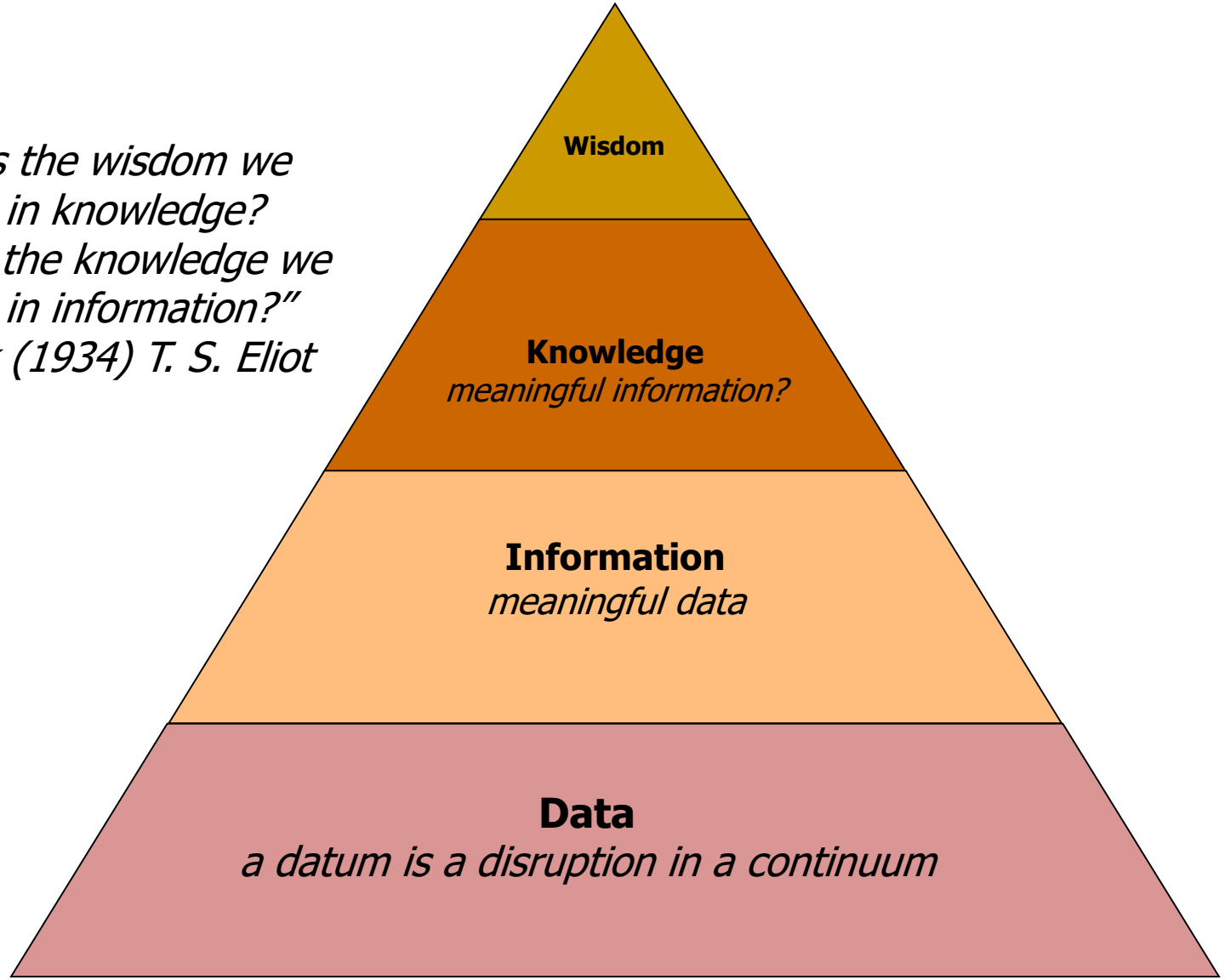
Informative annexes

- This is helpful information, you do not have to agree with it or apply it.

<u>The "purpose" of a standard is to be referenced in a contract</u>

Data and Information - the philosophical view

"Where is the wisdom we have lost in knowledge? Where is the knowledge we have lost in information?"
The Rock (1934) T. S. Eliot



Differentiating Information and Data

data: fixed form into which information is transformed so that it can be stored or moved.



Information → **Data** → **Information**

Copyright only covers "fixed form" (*you can only copyright data not information*)

Definition of data and information

ISO definitions (ISO/IEC 2382-1:1993)

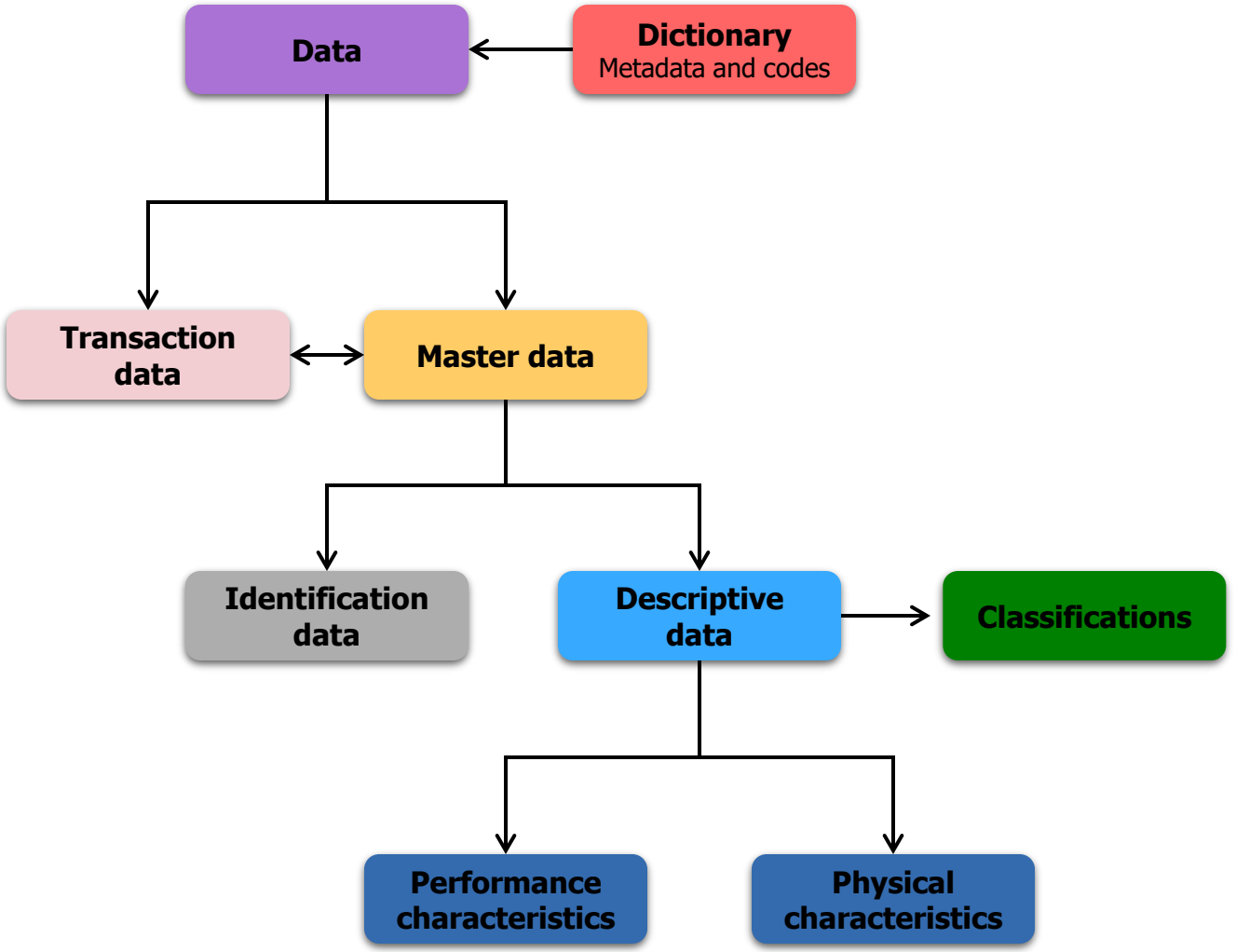
Information

knowledge concerning objects, such as facts, events, things, processes or ideas, including concepts, that within a certain context has a particular meaning

Data

Re-interpretable representation of information in a formalized manner suitable for communication, interpretation, or processing

Taxonomy of Data



Differentiating Information and Data

Information is Power

Timely
Relevant

Data is Truth

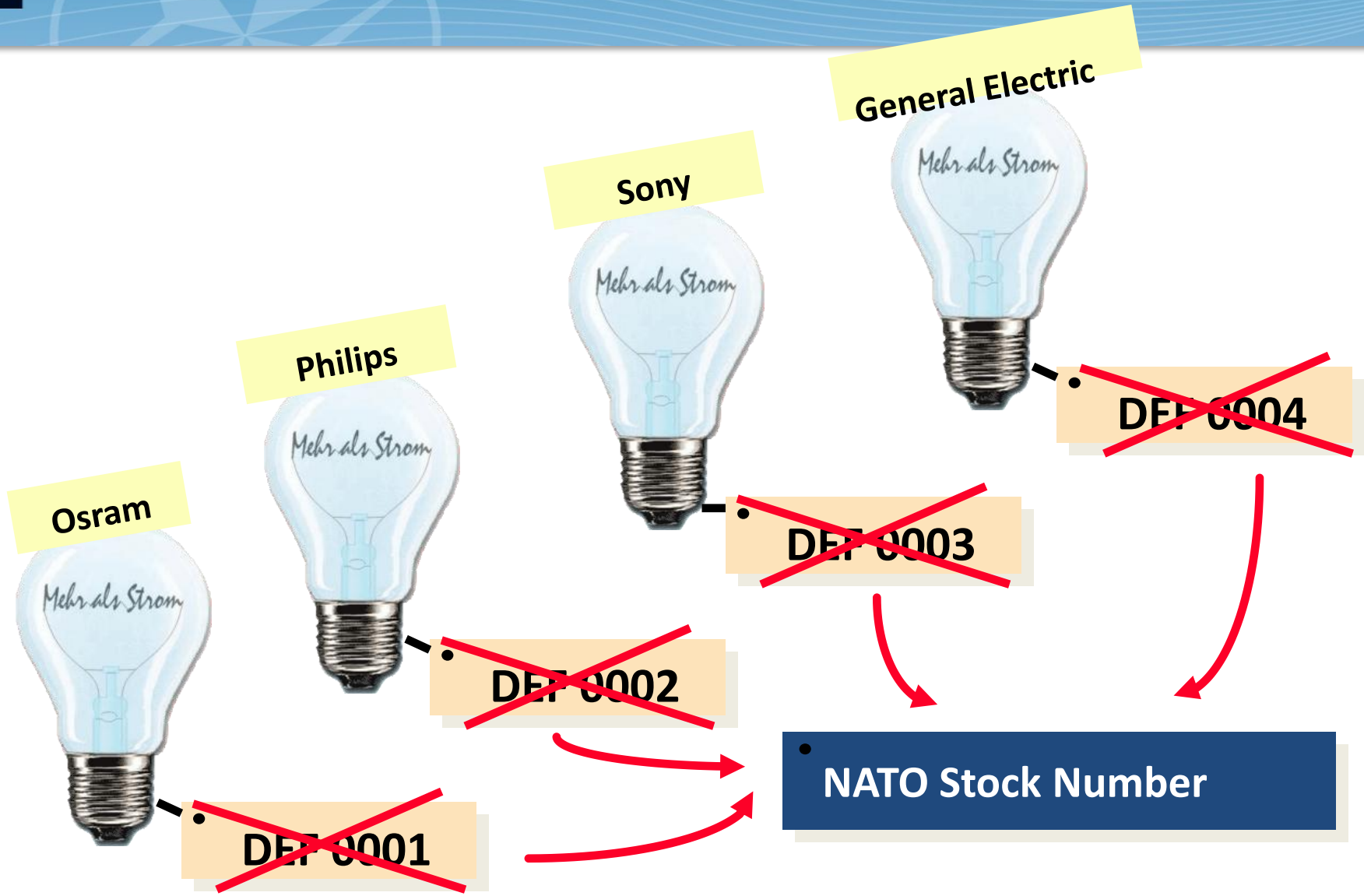
Unambiguous
Attributable

What is the Purpose of Codification?

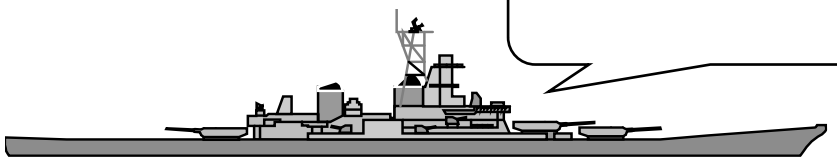
- To establish a common supply language throughout all logistic operations
- To enable interoperability
- To optimize resource management by minimizing duplication in inventories and allowing common management

The NATO Codification System (NCS)

What is Codification?

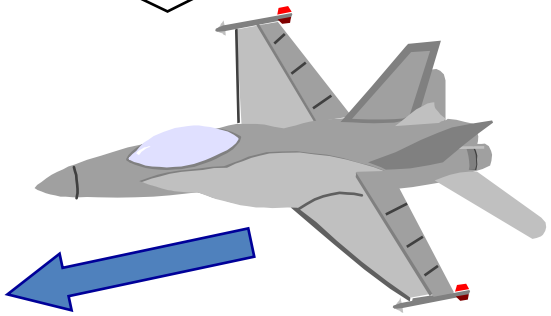


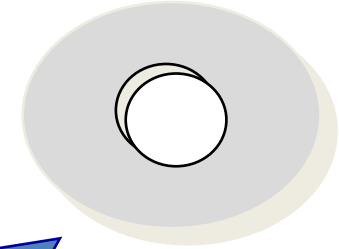
Before Codification



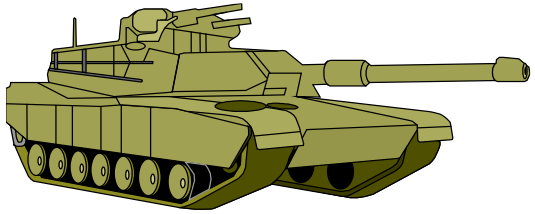
I need a washer

Jeg har brug for en skive





Eine Scheibe bitte



Une rondelle s.v.p.

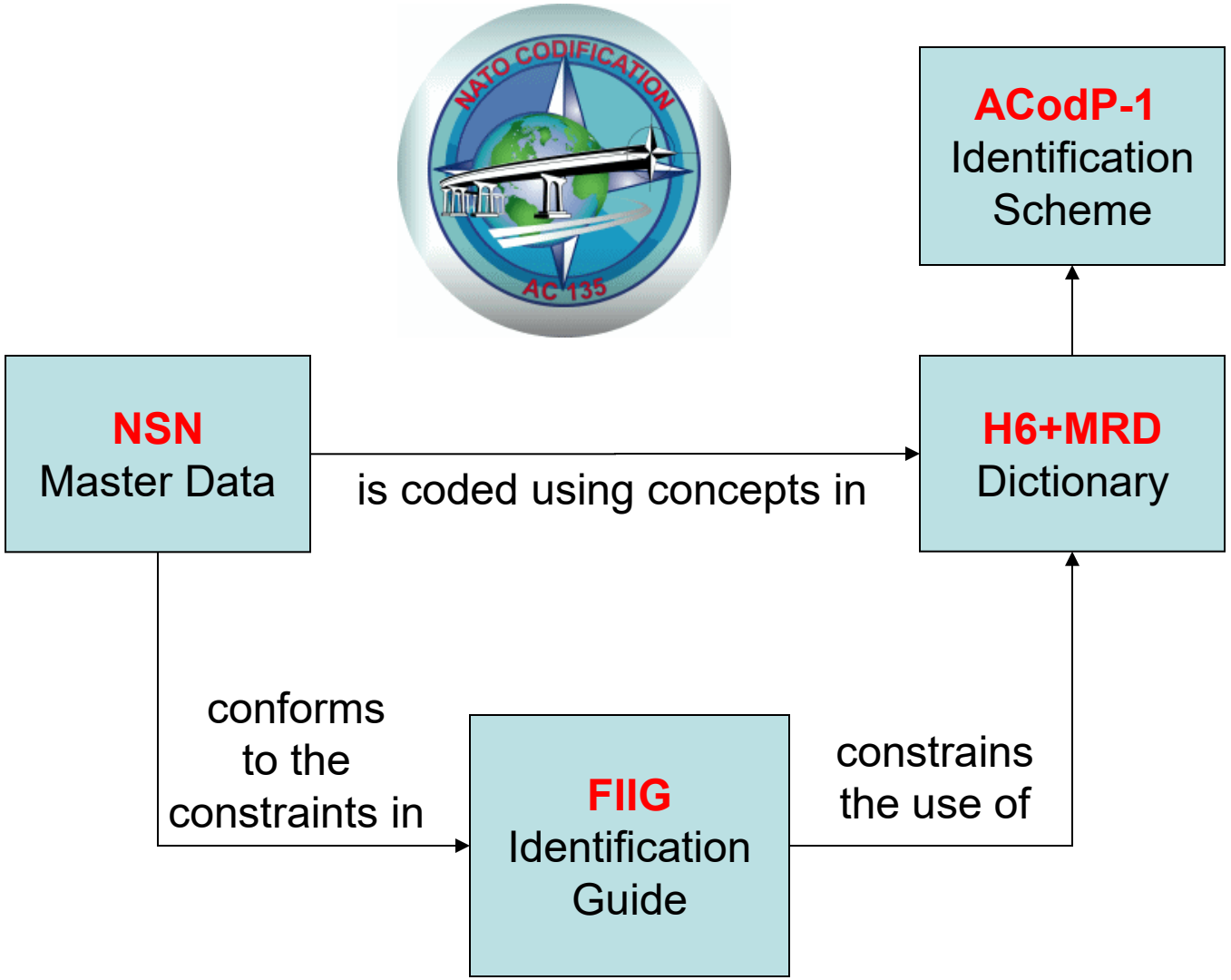


"There is and always has been a philosophical gulf between the application of cataloging for military purposes and … for commercial. …commercial practices are not precise enough to support cost-effective military inventory management and military cataloging is far too detailed and costly for commercial purposes …ECCMA offers a way to bridge the gulf" Mr. Alan Williams, Asst Dept Minister, Canadian Dept of National Defence

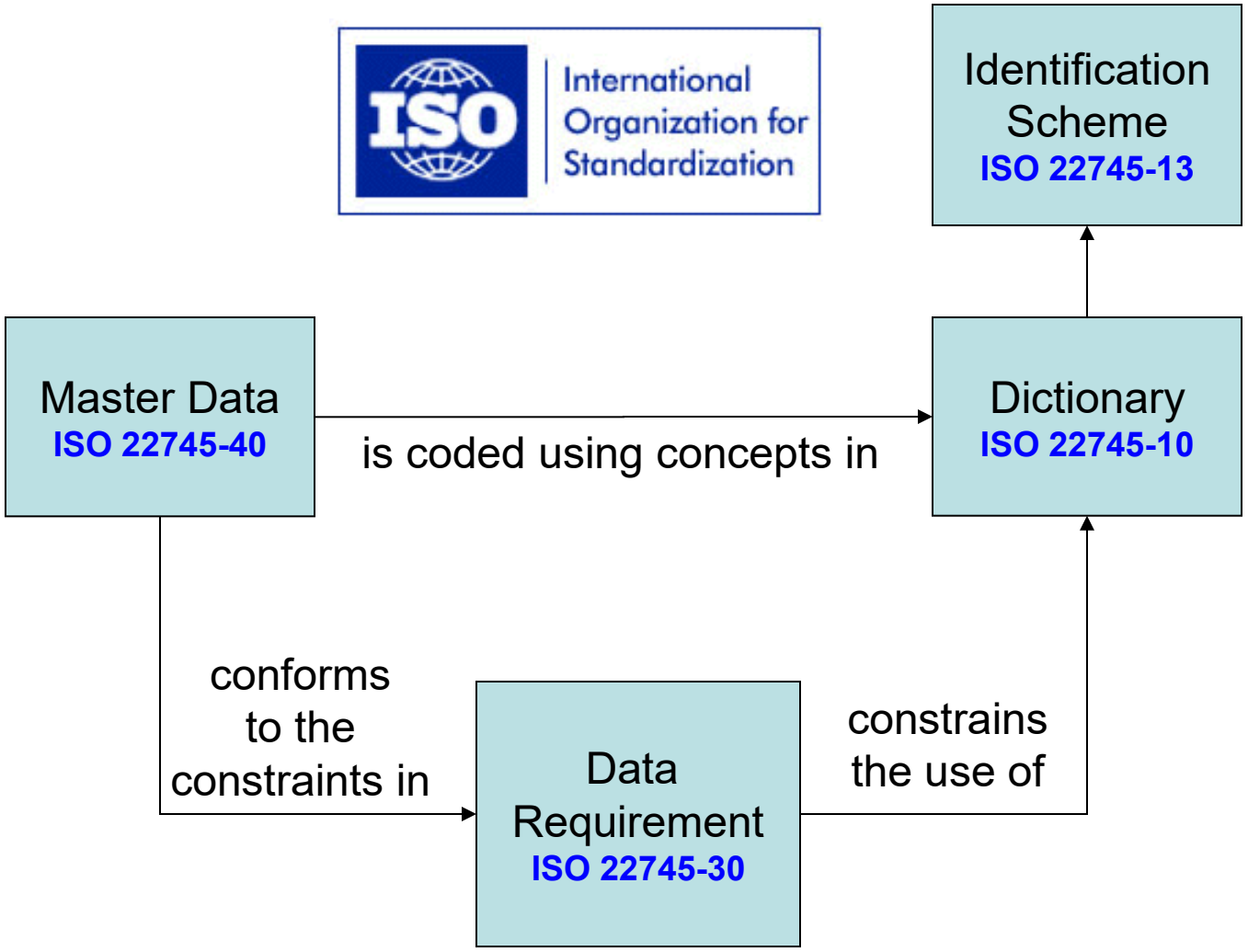


Portable Data is the Bridge to Interoperability

NCS Architecture



ISO 22745 Architecture



Motivation for ISO 22745 and ISO 8000

"From a logistics information perspective… an F-15 is just 171,000 parts flying in very close formation."

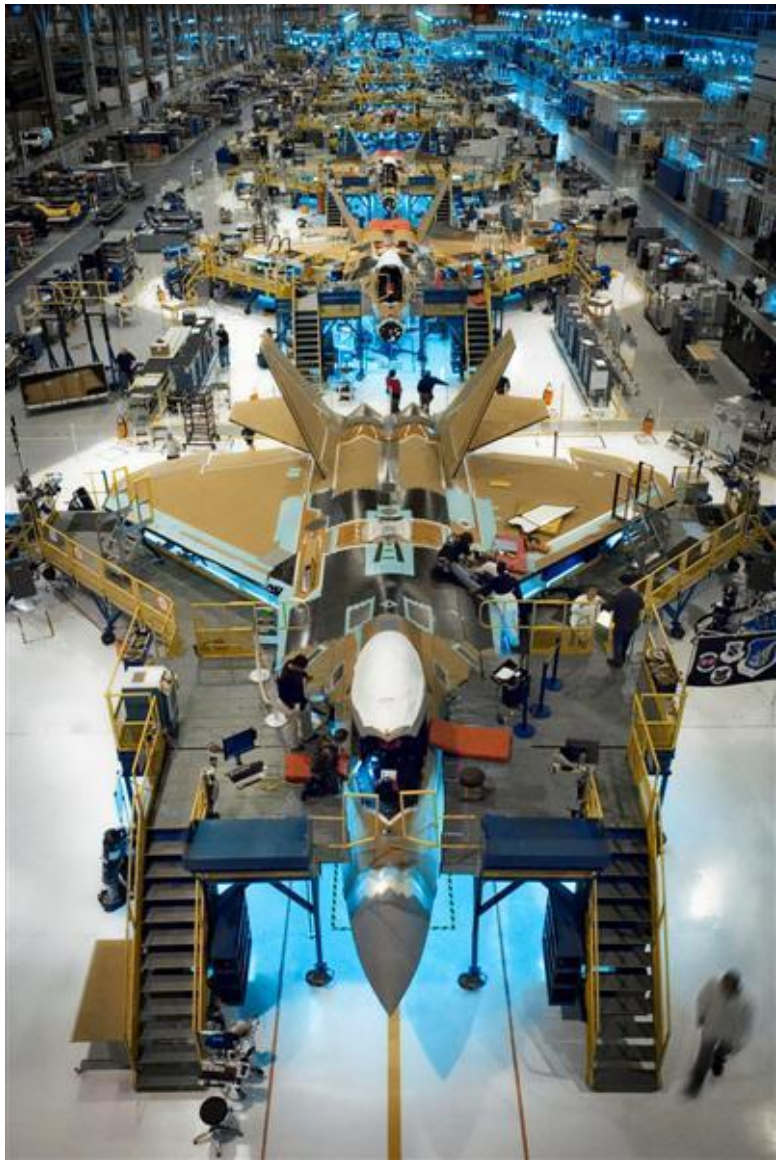


JULY 1st 1952 - Cataloging and Standardization Act, Public Law 82-436 as codified by United States Code, Title 10, Chapter 145 – Cataloging and Standardization Sec. 2451.

(a) The Secretary of Defense shall develop a single catalog system and related program of standardizing supplies for the Department of Defense.

(b) In cataloging, the Secretary shall name, describe, classify, and number each item recurrently used, bought, stocked, or distributed by the Department of Defense, so that only one distinctive combination of letters or numerals, or both, identifies the same item throughout the Department of Defense. Only one identification may be used for each item for all supply functions from purchase to final disposal in the field or other area.

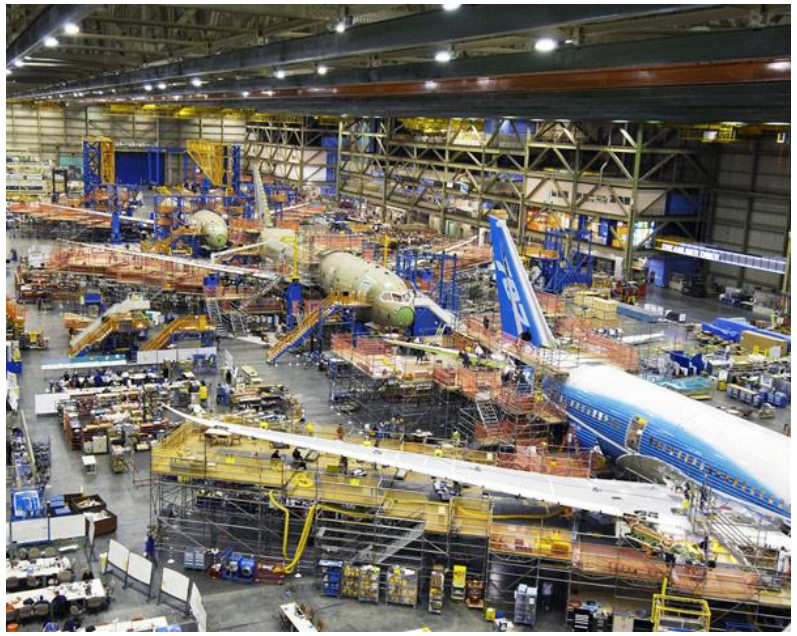
Motivation for ISO 22745 and ISO 8000





Controlling costs requires better asset, product , component and process visibility. This is achieved through faster, better and lower cost access to authoritative characteristic data.

Motivation for ISO 22745 and ISO 8000







Controlling costs requires better asset, product , component and process visibility. This is achieved through faster, better and lower cost access to authoritative characteristic data.

Quality ERP Descriptions

ANGLO COAL

52368965412 – Tire Bridgestone 435/95 R25

ANGLO BASE METALS

56329845 – Tyre BS 435/R25 Standard Purpose E3 2 Star Radial

ANGLO PLATINUM

125435 – Bridge Stone 25inch 435/95

KUMBA RESOURCES

965123465 – Tyre Bridgestone Part Number 12345



Standardised Long Description:
Tire: Pneumatic, Vehicular: Service Type for Which Designed: Loader Tire Rim Nominal Diameter: 25' Tire Width: 445mm Aspect Ratio: 0.95 Tire Ply Arrangement: Radial Ply Rating: 2* Tire & Rim Association Number: E3 Tread Material: Standard Tire Air Retention Method: Tubeless Tire Load Index and Speed Symbol: NA Tread Pattern: VHB TKPH Rating: 80

Standardised Short Description:
Tire Pneumatic: Loader 25‘ 445mm 0.95 2*

The eOTD is a foundation for design collaboration and industry standards.

ISO 22745 and the eOTD are the foundational enablers for the breakthrough our industry needs in the next generation of direct, accurate, and effective collaboration across the supply chain at meaningful and granular levels of data exchange never before imagined.

Alton Sanders
Senior Manager,
IDS Engineering Standards Control Function

PW Knowledge and Reuse Management (KARMA)



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Motivation for ISO 8000

Quality Master Data Provider view

Supplier and Manufacturers recognize that:

- data integration is one of the keys to a long-term relationship
- the ability to provide their customers with quality data is a significant differentiating factor.

Suppliers and Manufacturers are:

- publishing the specifications of their products, capabilities and services on their web sites.
- looking to increase their visibility and understand that the best way to do this is to improve the quality of their data.



Suppliers and manufacturers are looking for a Standard that they can use to differentiate the quality of their data.







When you order seafood from **Quality** Fresh Seafood, you can be confident that you are receiving the very best **quality** of seafood and delivery.









LABONAL, la chaussette **de qualité** depuis 1924



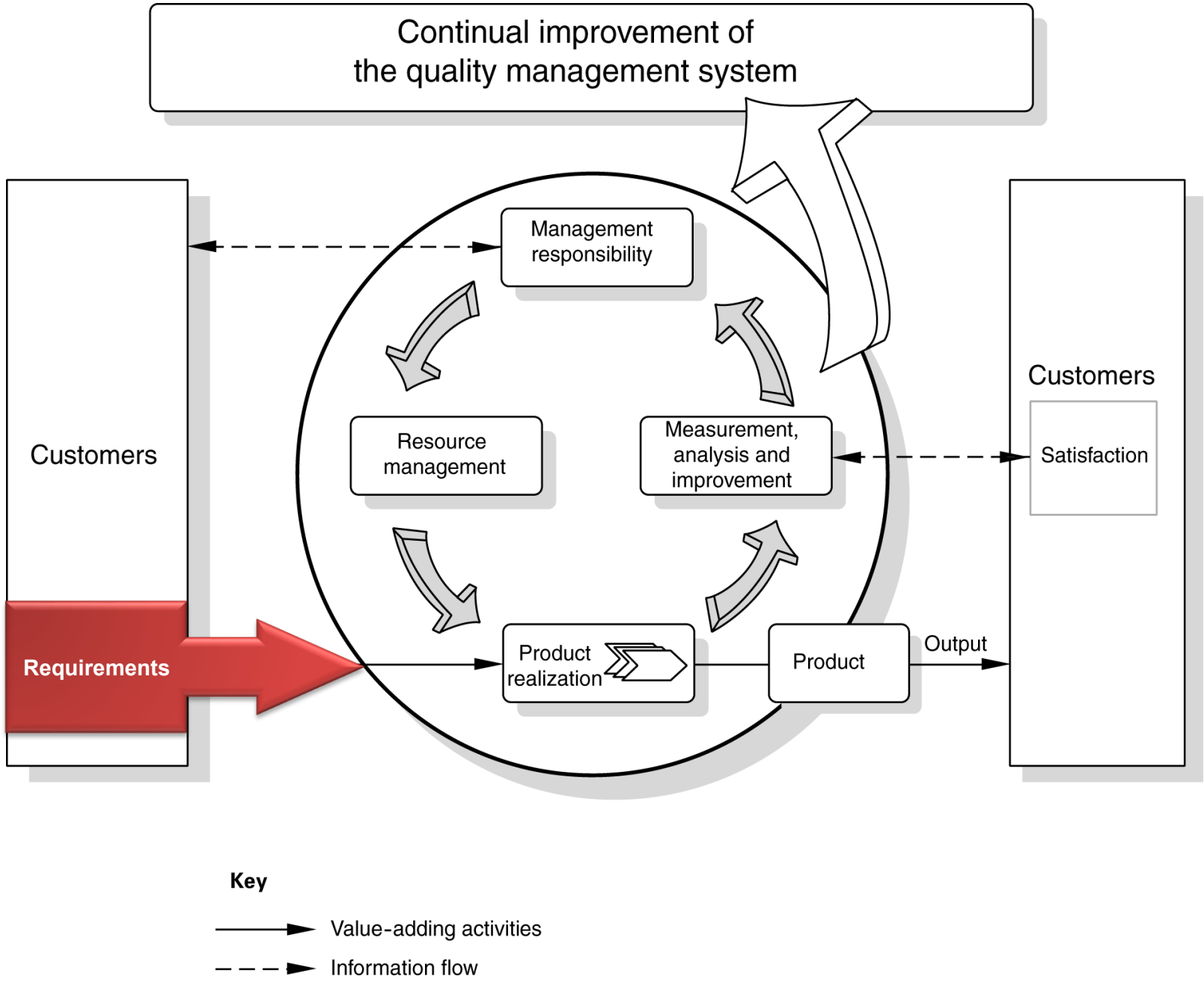
Annuaire **de qualité** pour être vu sur le web, vous propose une sélection des meilleurs sites internet.

Camping **Qualité** part d'une idée simple

a pour ambition de créer un réseau de blogs francophones **de qualité**

Les billards **de qualité** professionnelle
Tout joueur souhaite jouer sur une ardoise **de qualité**

ISO 9001:2000(E)



ISO 9000:2005(E)

3.1.1

quality

degree to which a set of inherent **characteristics fulfils requirements**

NOTE 1 The term “quality” can be used with adjectives such as poor, good or excellent.

NOTE 2 “Inherent”, as opposed to “assigned”, means existing in something, especially as a permanent characteristic.

3.1.2
requirement
need or expectation that is stated, generally implied or obligatory

NOTE 1 "Generally implied" means that it is custom or common practice for the **organization (3.3.1), its customers (3.3.5)** and other **interested parties (3.3.7), that the need or expectation under consideration is implied.**

NOTE 2 A qualifier can be used to denote a specific type of requirement, e.g. product requirement, quality management requirement, customer requirement.

NOTE 3 A specified requirement is one that is stated, for example in a **document (3.7.2).**

NOTE 4 Requirements can be generated by different **interested parties (3.3.7).**

NOTE 5 This definition differs from that provided in 3.12.1 of ISO/IEC Directives, Part 2:2004.
3.12.1
requirement
expression in the content of a document conveying criteria to be fulfilled if compliance with the document is to be claimed and from which no deviation is permitted

It is all about the requirements

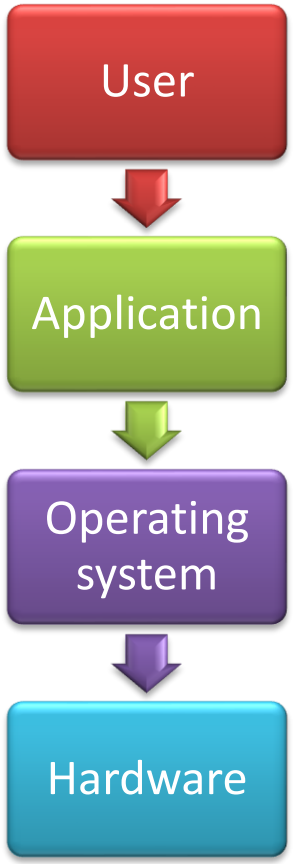
Quality data is data that meets requirements

nothing more – nothing less!

(well - maybe something more)

Quality data is portable data that meets stated requirements

Operating systems as a solution to hardware “lock-in”



An operating system is the infrastructure software component of a computer system responsible for sharing the limited resources of the computer. The operating system acts as a host for applications that are run on the machine.

1969 - UNIX
1974 - CP/M
1981 - MSDOS
1985 - Windows 1.0
1995 - BOB
2000 - ME
2001 - XP
2007 - VISTA
2009 - Windows 7
2009 - **Android**

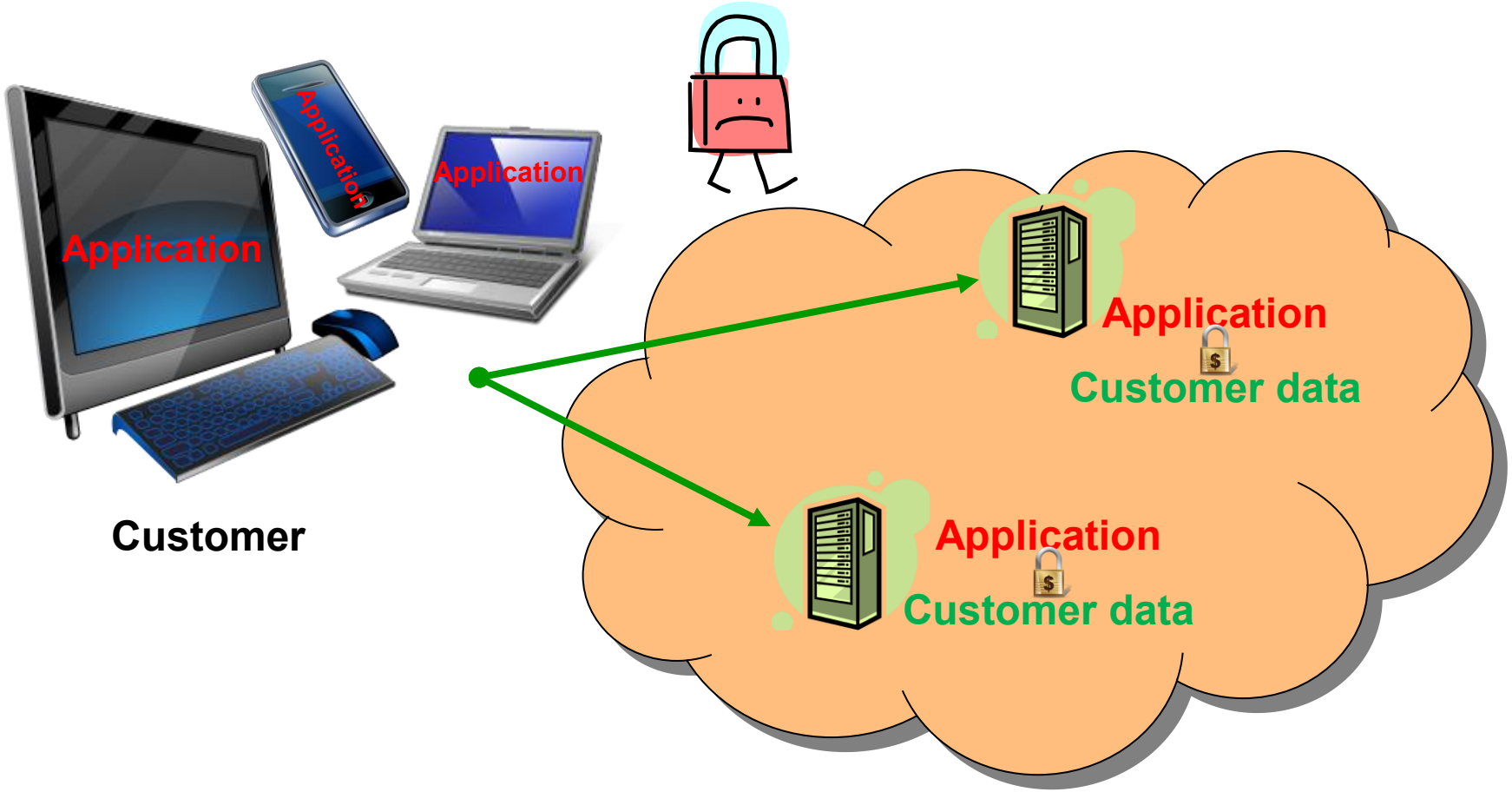
ISO data standards are the antidote to application “lock-in”



Quality data is <u>portable data</u>; it is independent of the software application and <u>accessible by any application.</u>

Vision of the Future

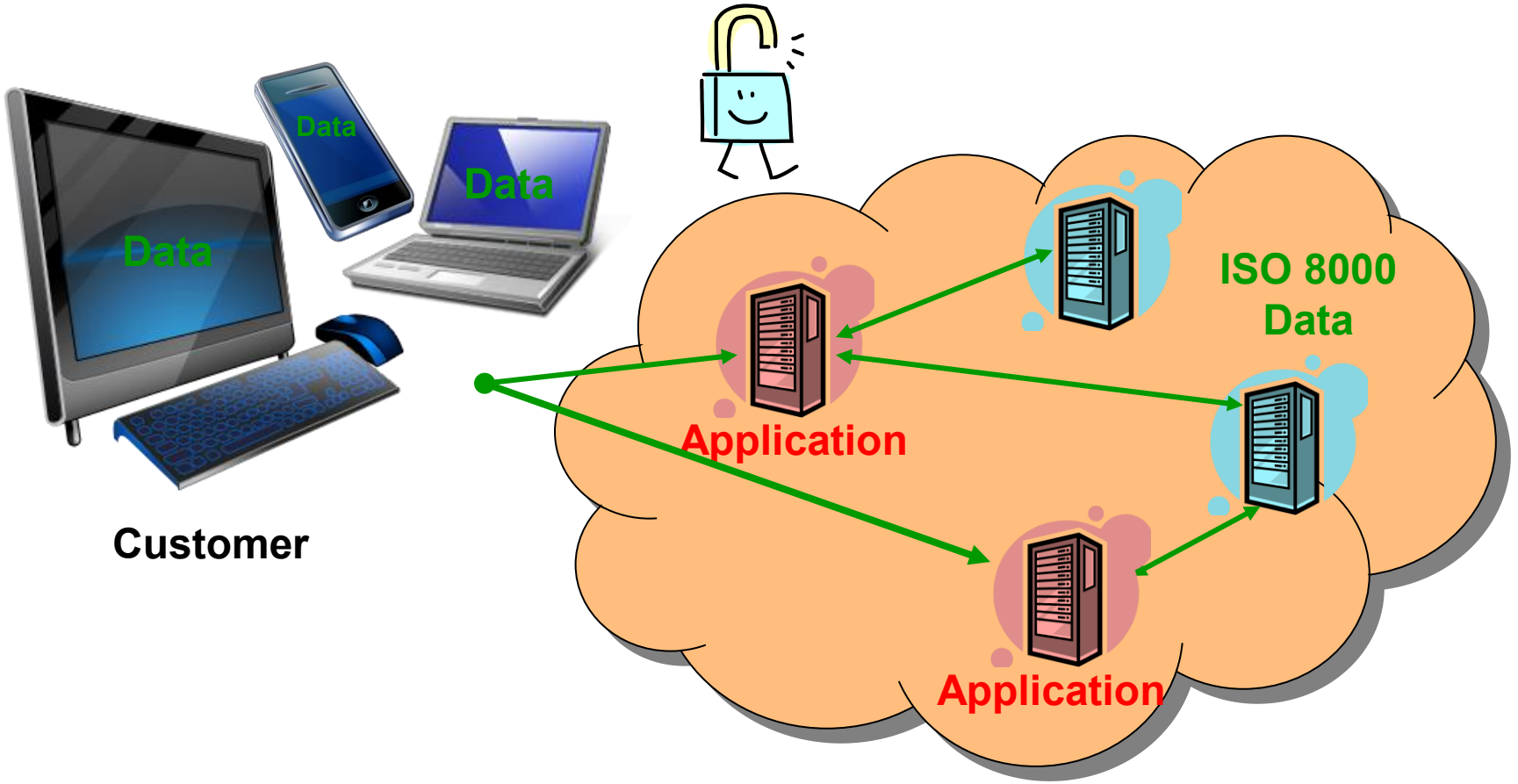
Service Provider View



Storing customer data <u>within</u> the application creates "customer lock-in"

Vision of the Future

Customer View



Data standards are the antidote to application "lock-in"

Property-Value pairs: The building blocks of meaningful data

Everything (Individuals, organizations, locations, goods, services, processes, rules, transactions) can be described using property-value pairs.

Property	Value
CLASS NAME	BOLT
PRODUCT NUMBER	3225020037
NOMINAL THREAD DIAMETER	1.0 INCH
WIDTH ACROSS FLATS	1.450 INCHES
WIDTH ACROSS CORNERS	1.653 INCHES
HEAD HEIGHT	0.591 INCHES
COUNT PER PACK	10
PACK PRICE	0.80 DOLLARS

Property	Value
FIRST NAME	JOE
LAST NAME	SMITH
TITLE	MANAGER
EMAIL	JOE.SMITH@COMANY.COM

Property	Value
DATE	2012-08-12
CREDIT CARD NUMBER	3225020037
MERCHANT NUMBER	15689345
AMOUNT	156.45 DOLLARS

Property	Value
STATE	PA
TITLE	33
CHAPTER	75
PARAGRAPH	3362
NAME	Maximum speed limits

Portable data is data where the metadata and reference data is defined

Everything (Individuals, organizations, locations, goods, services, processes, rules, transactions) can be described using property-value pairs.

Property	Value
CLASS NAME	BOLT
PRODUCT NUMBER	3225020037
NOMINAL THREAD DIAMETER	1.0 INCH
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Property	Value
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LAST NAME	SMITH
TITLE	MANAGER
EMAIL	JOE.SMITH@COMANY.COM

Property	Value
DATE	2012-08-12
CREDIT CARD NUMBER	3225020037
MERCHANT NUMBER	15689345
AMOUNT	156.45 DOLLARS

Property	Value
STATE	PA
TITLE	33
CHAPTER	75
PARAGRAPH	3362
NAME	Maximum speed limits

The Need for Concept Encoding



In South Africa a traffic light is called a robot!

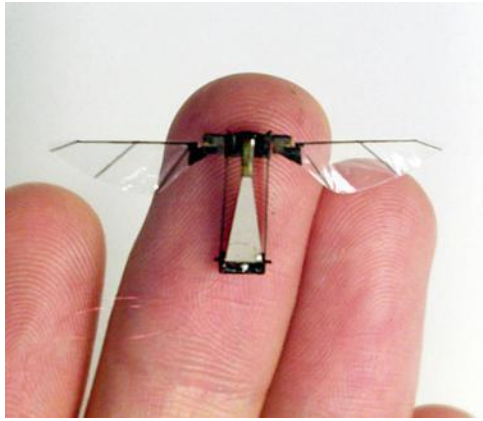




```
// opening a PR bot comment
app.on('pull_request.opened', async context => { // gfg
        const username = context.payload.pull_request.user.login
        const comment = context.issue({
            body: `Hey @${username} . Just so you know, you ROCK!`
        })
        return context.github.issues.createComment(comment)
    }) // code block ok


// welcome new contributors on their first PR
app.on('pull_request.opened', async context => { // gfg
        // username : pull_request.user.login
        // 2. use login to find count of PRs by user
        const listforrepo_result = await context.github.issues.listForRepo(context.repo({
            state: 'all',
            creator: context.payload.issue.user.login
        }))
        const number_PRs = listforrepo_result.data.filter(data => !data.issues)
        if (number_PRs.length === 1) {
            try {
                context.github.issues.createComment(context.issue({ body: 'first PR. sweet. Guess we owe you a
                coffee!' }))
                    //loc 2
            } catch (err) {
                if (err.code !== 404) {
                    throw err
                }
            }
        }
    }) // code block ok
```
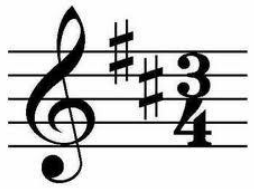
Robot



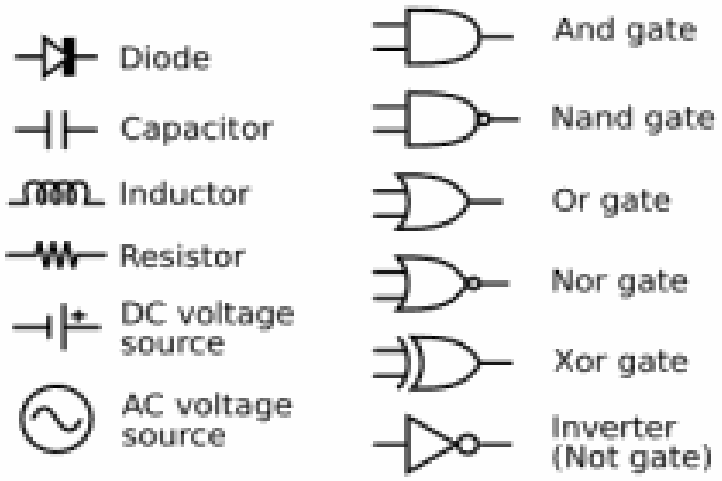
Concept Encoding

Just as with music notation and engineering symbols, a concept dictionary is used to assign identifiers to metadata and reference data concepts, and these are used to communicate more accurately in a language independent environment.

Music



Engineering



Concept dictionary

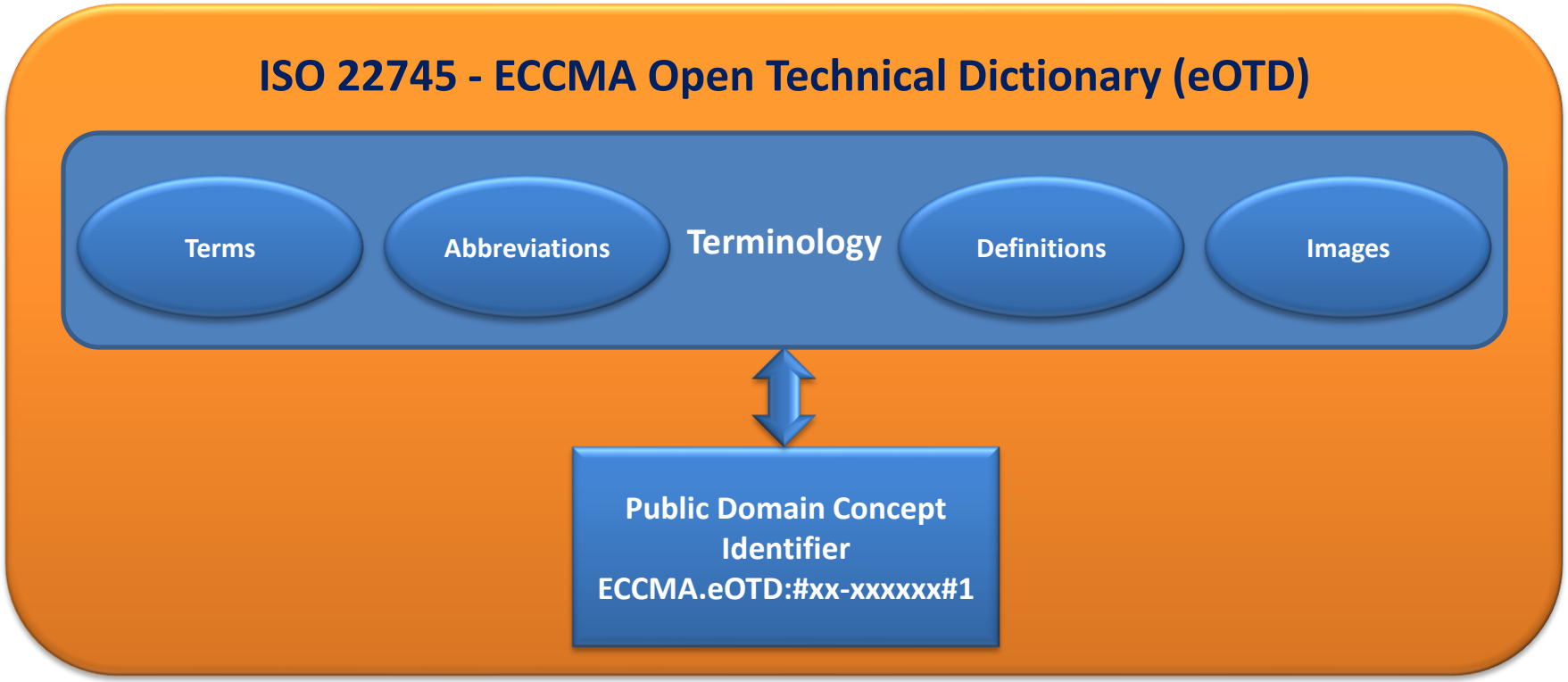
A unique public domain identifier is assigned to a concept.

ECCMA.eOTD:1#01-089388#1	**table**
ECCMA.eOTD:1#01-086445#1	**chair**
ECCMA.eOTD:1#02-018635#1	**weight**
ECCMA.eOTD:1#02-005808#1	**length**
ECCMA.eOTD:1#07-277660#1	**Monday**
ECCMA.eOTD:1#05-001122#1	**kilogram**

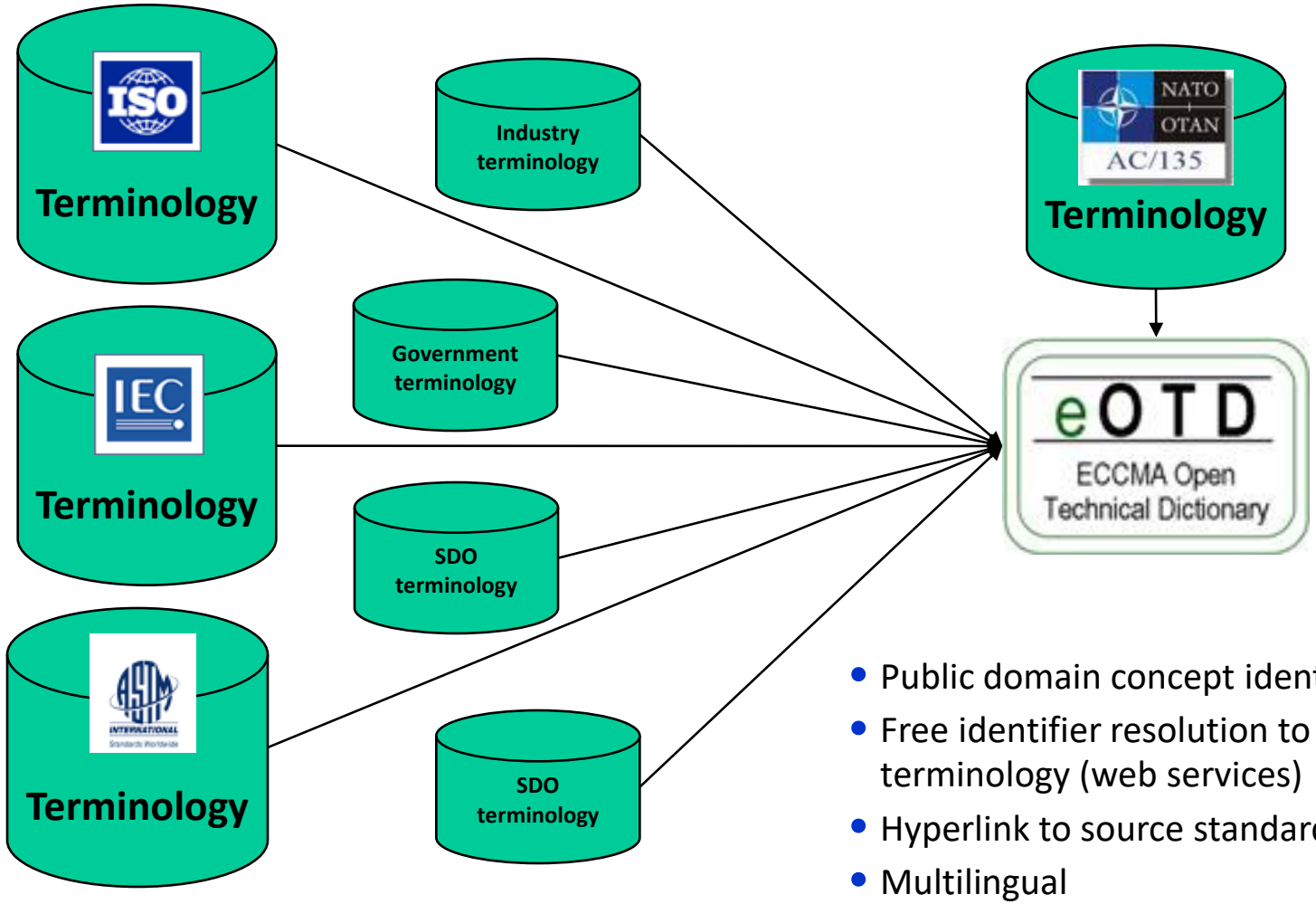
Publicly Visible Terminology in a Standard Model

The eOTD (ECCMA Open Technical Dictionary) is an ISO 22745-20 compliant central registry of terminology. Each concept and terminological component in the eOTD is assigned a unique and **permanent public domain identifier**.

Users create corporate preferred subsets of the eOTD and use the eOTD concept identifiers to manage **concept equivalence mapping** with the concepts used by their trading partners.



Publicly Visible Terminology in a Standard Model



- Public domain concept identifiers
- Free identifier resolution to underlying terminology (web services)
- Hyperlink to source standards
- Multilingual
- Multiple terms, definitions and images linked to single concept identifier

Applying Concept Equivalence to Enrich a Corporate Dictionary

SABIC

concept equivalence table

01-1073082	=	01-086142
01-1073082	=	01-068756

Sabic Bolt = Rockwell Bolt

Sabic Bolt = ASTM bolt

eOTD Concept ID	Term	Ex ref	Originating Organization	Definition	Status
0161-1#01-1073082#1	BOLT	IR237	SABIC	A fastener that is externally threaded on one end and generally with some style of head on the other end and is normally intended to be tightened or released by torquing a nut and designed to fasten objects together.	Active
Equivalent concepts					
0161-1#01-086142#1	BOLT	-	Rockwell Automation, Inc	A fastener consisting of a threaded pin or rod with a head at one end and designed to be inserted through holes in assembled parts and secured by a mated nut that is tightened by applying torque. Крепежная принадлежность, представляющая собой стержень с нарезанной резьбой и головкой на одной стороне, предназначенный для помещения в отверстия собираемых вместе деталей с последующей их фиксацией с помощью гайки, затягиваемой до определенного крутящего момента.	Active
0161-1#01-068756#1	bolt	F 1789 - F16	ASTM	headed and externally threaded fastener designed to be assembled with a nut	Active

ISO 25500

Unique Identification

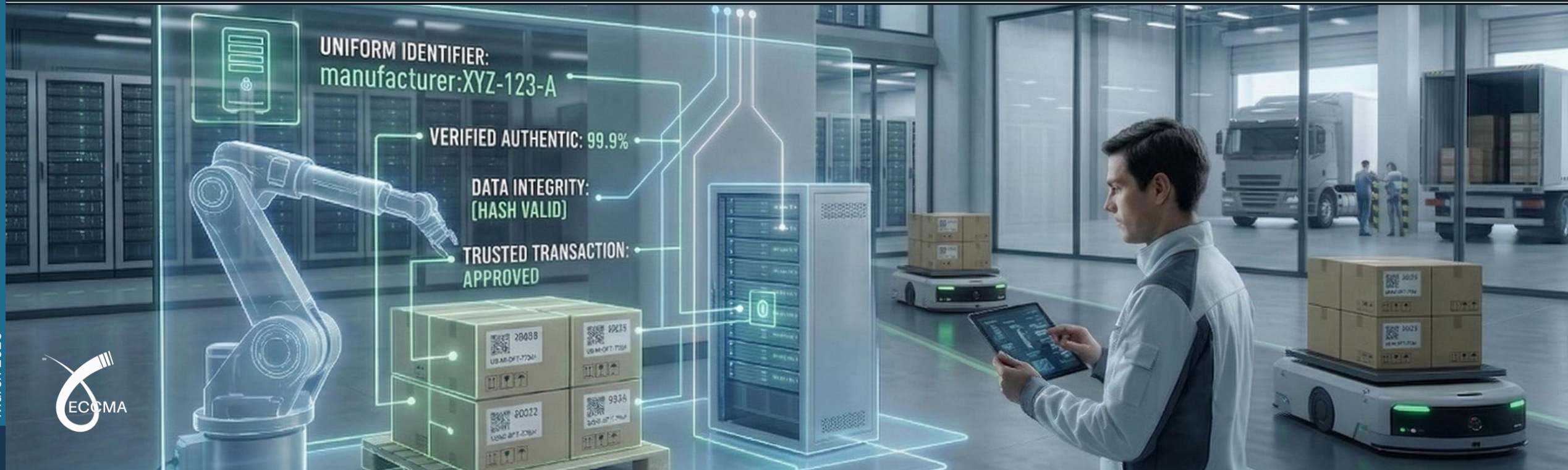
Verified Data

Interoperability

The Industrial Internet

Rebuilding supply chain trust through verified data and interoperability

No anonymity - No impersonation - No viruses



March 2026

What is the Industrial Internet?

A global operating environment — defined by ISO 25500 — where every company, product, and location can be uniquely identified and verified before supply chain transactions take place.

✓ No anonymity ✓ No impersonation ✓ No viruses

✓ AI has access to authoritative data ✓ Data owner controls who has access to their data

⚠ Not to be confused with IIoT (Industrial Internet of Things) — the Industrial Internet is about trust in supply chain data, not connected devices.

The Problem Today

Fragmented supply chains suffer from unverified data, fraud, delays, errors, and wasted resources — because trading partners cannot reliably trust each other's data.

The ISO 25500 Vision

A common language and structure where data about organizations, locations, and products or services is automatically verified against authoritative sources, controlled by the data owner, before business is transacted.

The Three Pillars of ISO 25500

1. Unique identification

- **WHO: Legal Entity Verification** — automated process to confirm legitimacy of trading partners
- **WHAT: International Part Number** (IPN) backed by technical data in Supply Chain Interoperable Format (scif)
- **WHERE: Natural Location Identifier** (NLI) — open-source geo locator for a loading dock or a location in a building (International Facility Identifier – IFI)

2. Data Verification

- **API Structure** — universal format for data verification requests and responses
- **Certificate Verification** — quality, safety, minority-owned, fair trade, environmental
- **Localization Data** — supports local sourcing priorities and provenance verification

3. Operating Procedures

- **Strategic Sourcing** — standardized data requirements for supplier selection
- **Procurement & Order Processing** — best practices for trusted transactions
- **Warehousing, Logistics & Returns** — end-to-end supply chain data standards

The Foundation: Unique Identification

ISO 25500 builds on ISO 8000 (the standard for industrial data quality). Before any data can be trusted, every entity, location, and item in the supply chain must be uniquely and unambiguously identified.

Legal Entity (IBID)

Every trading partner has an International Business Identifier — easily verified against official registries to confirm legitimacy before any transaction.

Item / Part Number

ISO 8000-115 defines the International Part Number format, so every product or service has a unique, machine-readable globally unique identifier that can be verified against the manufacturer.

Certificates & Attributes

Quality certifications, safety approvals, ethical sourcing credentials — all linked to verified entities and items, not just claimed on a document.

Authoritative Sources

Verification always traces back to the issuing authority — not a hard copy that can be forged. Machines can check automatically before a purchase order is placed and goods are routed.

Supply Chain Trust in the Age of AI

AI is a powerful accelerator — when built on verified data, it drives efficiency and automation. When built on unverified data, it amplifies risk at unprecedented scale.

Fraud Prevention	Counterfeit parts and fraudulent suppliers are blocked before orders are placed — AI procurement engines check IBID and part number against authoritative registries automatically.
AI Procurement Engines	ISO 25500-compliant open catalogs provide the clean, standardized data that AI engines need to match buyers with verified suppliers accurately and at scale.
Interoperability Across Borders	A universal API structure means any system — ERP, procurement platform, or AI agent — can request and receive verified supply chain data in a common format.
Sustainability & Compliance	Verified localization, environmental certificates, and labor practice data allow organizations and AI systems to act on ESG commitments, not just claims.

Who Benefits & How to Get Involved

Businesses

- Reduce procurement fraud and counterfeit risk
- Lower transaction costs through automation
- Improve ESG compliance and reporting

Governments

- Greater supply chain transparency
- Tools for enforcing local content rules
- Support for trade policy and sustainability goals

Consumers

- Confidence in product origin and quality
- Visibility into ethical sourcing practices
- Trust in certifications and labeling

How to Engage with ISO 25500

01 Understand ISO 8000

Master the data quality foundation — unique identification of entities and items.

02 Adopt Open Catalogs

Use ISO 25500-compliant standardized open catalogs to feed AI procurement engines.

03 Verify Trading Partners

Implement IBID-based legal entity verification before onboarding suppliers.

04 Join the Ecosystem

Engage with ECCMA and ISO to shape and implement the standards.

How to Access the Industrial Internet

Just as you need an Internet Service Provider (ISP) to access the Internet, you need an Industrial Internet Service Provider (IISP) to access the Industrial Internet.

Internet (familiar)

- Requires an Internet Service Provider (ISP)
- ISP gives you access, routing, and an address
- Without an ISP, you cannot connect to the internet

Industrial Internet

- Requires an Industrial Internet Service Provider (IISP)
- IISP gives you verified identity, data access, and interoperability
- Without an IISP, you cannot participate in the Industrial Internet

VDSAI.net — The World's First IISP

VDSAI.net is the first Industrial Internet Service Provider. By subscribing to VDSAI.net, businesses gain access to the Industrial Internet — enabling verified identity, trusted supply chain data exchange, and full ISO 25500 compliance.

Resources – The Industrial Internet and ISO 25500

Subscribe to
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Lead the way, join the Industrial Internet community

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Learn more…

Access the ISO 25500 standards

Wiki “Industrial Internet”

Peter R Benson | Executive Director
ISO.IBID:**US-DE.BER:3031657|Code Management Association|19990420**
Supply chain data quality and interoperability through International Standards
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ISO 25500 International Reference

For example, used to format a local part number as an an International Part Number (IPN)

Corning.LS:4537



ISO 25500 International Business Identifier (IBID)

ISO.IBID:US-UT.BER:7252316-0160|I5 SERVICES, LLC|20090128|D

- ISO domain
- ISO 25500-3 registered Jurisdiction Prefix
- Authoritative Legal Entity Identifier (ALEI)
- Legal Name
- Date of Origin
- Citizenship **D**omestic or **F**oreign

IBRN

IBID

the first line of defense against corporate identity theft (and duplicate supplier records)

The most common source of corporate fraud is identity theft where a bad actor registers a business under the same name as a known business but in a different jurisdiction (and on a different date). Knowing the IBID of the legitimate business protects against this type of fraud.



While IBID verification confirms legal identity, a business may have a status of "inactive" if it has not met the Registry requirements - for example non-payment of annual registration fees.

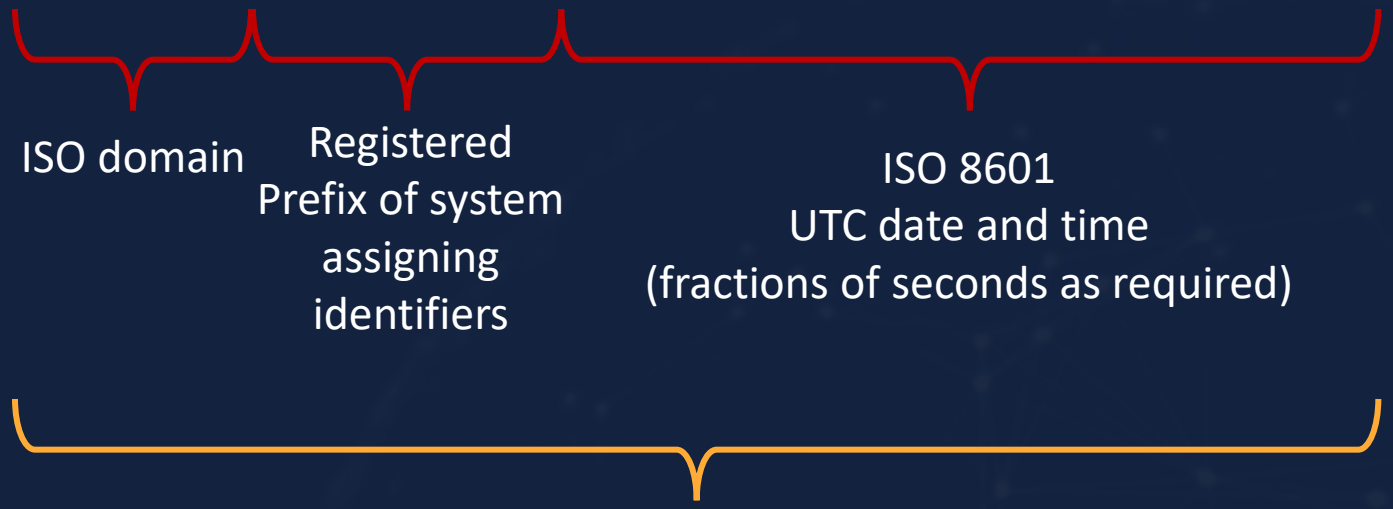
ISO 25500-120 International Facility Identifier (IFI)

ISO.IFI:US-DE.BER:3031657|ISO.NLI:94MNRT-5AC9U1-H0



ISO 25500-4 International Transaction Identifier (ITI)

ISO.ITI:Corning|2026-06-30T18:43:08.99



Supply Chain Data Formatting Contract Clause





Supplier hereby accepts responsibility for providing the data necessary for **the installation, operation, maintenance, or training purposes** of the item covered by this contract including studies, analyses, test data, or similar data, when the study, analysis, test, or similar work was specified as an element of performance; specifically, the data shall be formatted as follows:

1. All dates and time shall be formatted in accordance with ISO 8601.
2. All parties supplying items (goods or services) under this contract shall be identified by their International Business Identifier (IBID) formatted in compliance with ISO 25500-3.
3. All facilities shall be identified using an International Facility Identifier (IFI) formatted in compliance with ISO 25500-120.
4. All items shall be identified using a reference formatted in compliance with ISO 25500-4.
5. The technical data that describes the physical, functional, and performance characteristics of an item, component, or process shall be provided in Supply Chain Interoperable Format (.SCIF) in compliance with ISO 25500-5.
6. All transactions shall be identified using international transaction identifier in compliance with ISO 25500-4



Supply Chain Data Formatting Contract Clause
ADOPTION DRIVERS



CUSTOMS

Automated validation
of Harmonized Tariff System (HTS) codes assigned by exporters and importers, using ISO 25500-4 formatted international part numbers and their links to ISO 25500-5 formatted specifications

- **Use AI to verify accuracy of HS code assignment**



GOVERNMENT

Local content policies and Non-Tariff Measures
national adoption of ISO 25500 formatted international part numbers and standardized specifications both for local products and for imported products that compete with local products as well as for technical standards allows automation of local content compliance verification



APPLICATION PROVIDERS

Interoperable data

Increasingly important to users looking to avoid software application lock-in

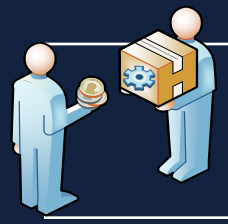
Master Data Management (MDM), Procurement automation or Product Information Management (PIM) applications that comply with ISO 25500-5 are seen to protect the integrity of their customer's data



ISO 25500-5 interoperable data formatting is trivial from a technological point of view.

Supply Chain Data Formatting Contract Clause

ADOPTION DRIVERS



SUPPLIER

- Improved product visibility in an AI environment
- Controlled distribution of data
- Easier to comply with buyer onboarding requirements





BUYER

- Lowers the cost of data acquisition
- Improves the accuracy of AI
- Easier onboarding of suppliers and items



www.eccma.org

www.iso.org

ISO Standards for
Data Quality and Unambiguous Exchange
and
Supply Chain Interoperability and Integration

- Brief Summary of Parts for ISO 22745, ISO 29002, ISO 8000, and ISO 25500

Standard	Title	ISO Committee
ISO 22745	Industrial automation systems and integration — Open technical dictionaries and their application to master data	ISO/TC 184/SC 4 Industrial Data
ISO 29002	Industrial automation systems and integration — Exchange of characteristic data	ISO/TC 184/SC 4 Industrial Data
ISO 8000	Data Quality	ISO/TC 184/SC 4 Industrial Data
ISO 25500	Supply Chain Interoperability and Integration	ISO/TC 184/SC 5 Interoperability, integration, and architectures for enterprise systems and automation applications